# Project: Local File Transfer (Laptop ↔ Phone)

## Objective

Build a modern local file transfer application that allows users to transfer files between a laptop and a mobile phone over the same local Wi-Fi network without using the internet.

The application must work entirely on the Local Area Network (LAN). No cloud storage, external servers, or internet connectivity should be required after installation.

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# Technology Stack

Backend:

- Node.js
- Express.js

Frontend:

- React
- Tailwind CSS

Optional:

- Socket.io for live transfer progress
- Multer for uploads
- QR Code generation
- mDNS/Bonjour for device discovery

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# Functional Requirements

## 1. Dashboard

Show:

- Device Name
- Local IP Address
- Connection Status
- Connected Device Name
- Current Transfer Speed
- Files Sent

- Files Received

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## 2. Upload

User should be able to

- Drag & Drop files
- Select files
- Upload folders
- Upload multiple files

Supported:

- Images
- Videos
- Documents
- ZIP
- APK
- Any file type

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## 3. Download

The receiver should see

- File name
- Size
- Type
- Sender

Buttons

- Download
- Download All
- Reject

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## 4. QR Connection

Generate QR code containing

http://<local-ip>:<port>

Phone scans QR.

Browser opens automatically.

No typing required.

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## 5. Mobile Interface

Responsive UI.

Pages:

Home

Receive Files

Send Files

Transfer History

Settings

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## 6. Desktop Interface

Sidebar

Home

Transfer

History

Settings

About

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## 7. Transfer Progress

Show

Percentage

Current Speed

Estimated Remaining Time

Transferred Bytes

Pause

Resume

Cancel

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## 8. Transfer History

Maintain local history.

Show

Date

File

Sender

Receiver

Size

Status

Allow clearing history.

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## 9. Device Discovery

Automatically detect devices connected to same LAN.

Display

Device Name

IP

Platform

Online Status

Ping

Connect button

Fallback to manual IP entry.

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## 10. Security

Generate one-time pairing code.

OR

Allow password protection.

User must approve every incoming transfer.

No internet communication.

No analytics.

No telemetry.

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## 11. Settings

Transfer folder

Theme

Dark/Light

Auto accept

Auto save

Maximum upload size

Port number

Device name

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## 12. Notifications

Desktop notification

Transfer started

Completed

Cancelled

Failed

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## 13. File Preview

Images

Videos

PDF

Audio

before downloading.

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## 14. Performance

Support

10GB+

Multiple simultaneous transfers

Resume interrupted transfers

Chunked upload

Streaming

Low memory usage

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## 15. APIs

POST /upload

GET /files

GET /download/:id

DELETE /history

GET /devices

POST /pair

GET /status

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## 16. Architecture

Backend

Controllers

Routes

Services

Middleware

Utilities

Frontend

Components

Pages

Hooks

Services

Store

Assets

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## 17. Nice UI

Use modern glassmorphism.

Rounded cards.

Animations.

Dark mode.

Mobile-first.

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## 18. Extra Features

Search files

Recent transfers

Favorite devices

Copy IP

Generate QR

Share text

Clipboard sharing

Send screenshots

Folder transfer

File compression

Transfer speed graph

Bandwidth limiter

Transfer queue

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## 19. Code Quality

Use TypeScript if possible.

Well-commented code.

Reusable components.

Environment variables.

Error handling.

Validation.

Logging.

Production-ready.

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## 20. Deliverables

Generate

- Complete backend
- Complete frontend
- Folder structure
- README
- Installation guide
- API documentation
- User manual
- Docker support
- Build scripts
- Windows executable instructions
- Linux instructions
- Mac instructions